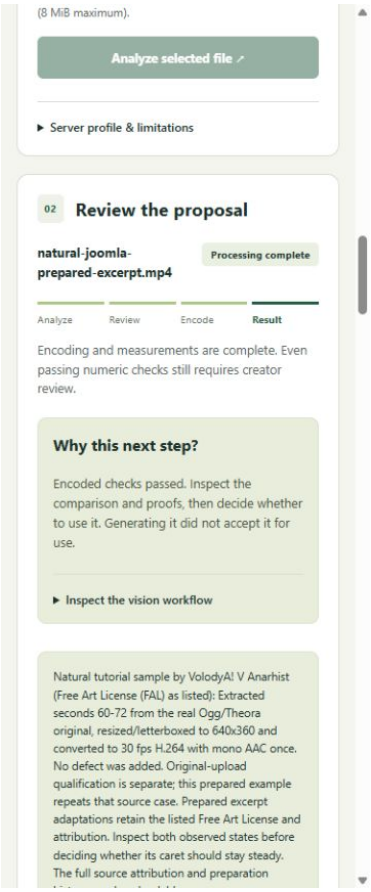
Frame Review / Keep the Lesson

Calm a reading pause. Keep the changing explanation.

A teacher has a finished recording and no editable project. A tiny idle caret blinks beside a participant number or changing command output. Frame Review proposes a bounded hold, asks the creator to protect teaching areas and approve a fresh source-bound plan, then checks the actual encoded result. The original stays recoverable.

The new “Why this next step?” panel explains the workflow: keep source, request human review, wait for tools, inspect a comparison or withhold a recommendation. Its owned JSON snapshot binds source, prediction, recipe and reported gates, and enters the rescue pack. It is a read-only state explanation, not an invented agent transcript or acceptance for use.



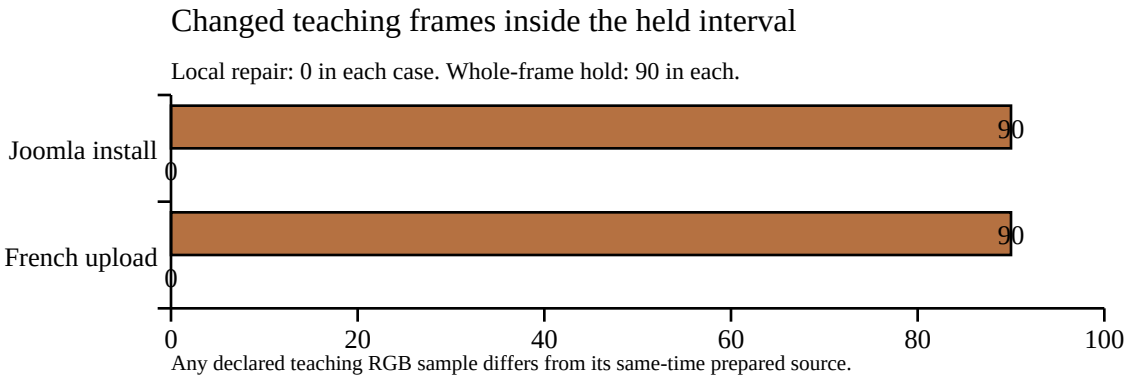
Actual live Brave Joomla workflow at 375 pixels. Fresh consent was unchecked and generation disabled before review. The explicit plan held brightness from observed frame 174 in a 96-pixel caret box over frames 114–209, with three named teaching areas. After generation, all 30 source-bound boundary/teaching PNGs loaded. Width checks also pass at 890.

Source: VolodyA! V Anarhist, Joomla Installation, Free Art License as listed. Seconds 60–72 prepared/adapted. No defect added or endorsement. Exact source/license URLs and hashes remain in the reproduction document.

Current qualification: all 14 Linux stages in AWS CodeBuild, 137 backend tests and 119 studio tests (one skip). Both selected untouched originals pass startup preflight and independent native/RGB/audio/timing/teaching/cut/tamper checks. Current AWS HTTPS API uploads, reviewed approval, encoding, complete 51/45-file packs and owned workflow downloads pass; every media/proof PNG byte matches native qualification. These are source regressions, not new discoveries.

Why activity reduction needs a preservation contract

Independent encoded video-only ablation: keep source vs reviewed local hold vs a deliberately constructed whole-frame hold, using the same interval and observed reference. Both edits reduce selected inside-hold activity to zero. Only the local repair leaves the declared teaching frames unchanged. This is an engineering comparison, not a teacher preference or learning result.



Case / source → local	Inside hold	Whole excerpt	Local / whole hold changed teaching frames
French upload assistant	5 → 0	8 → 2	0 / 90 of 97
Joomla installation	4 → 0	16 → 10	0 / 90 of 96

Activity counts adjacent selected-mean RGB changes ≥8 codes; teaching differences include codec variation, not semantic updates. The counterfactual passes an all-frame native-sample roundtrip; source/local teaching digests match pinned Linux results. Initial unverified-color baseline numbers were excluded and the encoder corrected before comparison. A Windows FFmpeg 9 digest mismatch remains recorded; these accepted numbers use pinned Debian FFmpeg 7.1.5.

Architecture: creator upload → bounded preparation on ECS → OpenCV 5 differences/components and source-matched proposal → creator teaching plan/approval → observed-luma FFmpeg edit with original chroma → encoded/native/RGB/audio/timing/cut gates → comparison and offline rescue pack. Failed processing/checks remain distinct from no proposal or a creator choosing unchanged.

AWS default-lesson-rescue:18 completed with one 1-vCPU/2-GiB task, one worker/eight jobs, owned cookies, CSRF/origin checks, bounds and 24h expiry (restart may clear sooner). No application task role or larger production resource was added. Temporary scoped CodeBuild access was removed after qualification. No COOL or special-award eligibility claim.

Image: sha256:dd6f6b6cdb4e3cc7ca624bc7ab761465566f66361a3f2539910a18cade5be7e6
Runtime manifest: 87337c7b9caa164b741ede5f35d4fc471d6110ac4e985935c2651f6b883d51f1 (119 files). App layer over the exact v11 image, not a cold dependency rebuild. Detector, encoder and preservation thresholds unchanged.

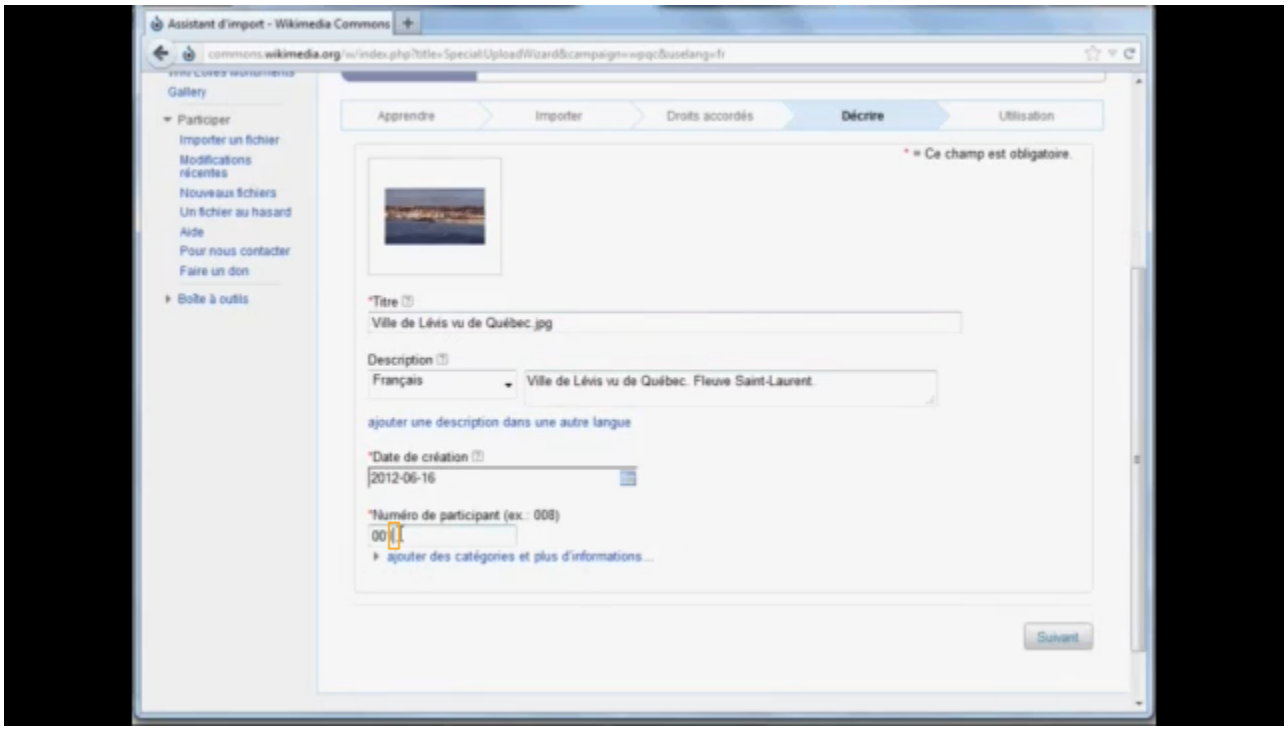
Reproduce: docs/workflow-v12/REPRODUCE.md, pinned runtime and root Dockerfile; python studio/run.py --verify /tmp/new-gate. Original qualifiers, exact cases, ablation script and machine-readable receipts included. The rescue pack retains review.html, every proof PNG, source/recipe/report and vision-workflow.json under its file-hash manifest.

Limits: selected 1–12-second SDR excerpts, originals ≤64 MiB; preparation changes picture/audio and can reduce label readability. Exact checks reference prepared bytes, not original resolution. No full-lesson splice-back or semantic tracking. Native CodeBuild preflight passed; prior WSL/preparation failures remain historical and do not establish general cold-start reliability. Browser ZIP completion callbacks remain unqualified; API downloads and browser UI/rendering are separate. Teacher preference, time saved, learning, attention and accessibility remain unmeasured. Existing video unchanged; new demo awaits user audio.

Frame Review / Keep the number. Calm the caret.

An edit should keep its promised footprint.

A real French Commons upload lesson shows a participant-number field. Its idle caret blinks beside the digits. Our first export passed the declared teaching boxes, but an extra check caught changed number pixels at frame 246. We rejected it. The final export holds only observed brightness samples inside the caret box, retaining every original color sample on its own frame. It now requires exact decoded RGB outside the actual box, including the earlier chroma influence margin. No existing teaching, audio, timing or cut gate was relaxed.



Static source frame with candidate overlay. Benoit Rochon, Tutoriel de l'assistant d'import, CC BY 3.0 as listed by Commons. Excerpt 140-152 seconds, prepared 640x360/30fps. Source image adaptation; author/license/original hashes and URLs are included. No added defect, endorsement or independent rights-clearance claim.

Current result: caret activity 5 to 0 inside the hold, 8 to 2 over the full excerpt, zero newly timed threshold events. Every one of 360 teaching RGB frames and the extra participant-number rectangle match exactly. All original chroma samples and RGB outside the 2x10 caret box remain exact; prepared audio, timestamps and outside-interval frames pass. Encoded selected/influence entry, return and loop comparisons do not worsen. Counts mean adjacent ROI-mean RGB changes of at least 8 codes; they do not measure learning benefit.

Earlier failed mapping and luma-build timeouts remain separate from the pack run. Its first French preparation and two Gentoo imports timed out; isolated runs passed with unchanged limits. Four initial corruption tests encountered the explicit transfer-busy 409; the qualifier now waits for that busy state before requiring strict refusal. Cold-start reliability is not established. Mapping: observed-luma-source-chroma-v2. Small source-color variation may remain inside the caret.

Source qualifications and retained failures

Original-source case	Hold activity	Whole excerpt	All checks
gentoo-equery	4 -> 0	13 -> 9	Pass
joomla	4 -> 0	16 -> 10	Pass
jupyter	4 -> 0	24 -> 18	Pass
libreoffice-writer	6 -> 0	15 -> 9	Pass
npm-basics	5 -> 0	17 -> 11	Pass
npm-packages	5 -> 0	19 -> 13	Pass
terminal	4 -> 0	15 -> 9	Pass
upload-assistant	5 -> 0	8 -> 2	Pass
wiki-citations	4 -> 0	6 -> 2	Pass
wiki-userpage	5 -> 0	17 -> 11	Pass
wikilink	4 -> 0	9 -> 3	Pass

11 selected untouched-original export cases pass on the final mapping. The French source is new; Joomla and the earlier nine are regression repeats, not eleven new lessons or independent creators. Every case includes actual HTTP preparation/approval/artifacts plus independent native/RGB/audio/timing/cut checks and teaching-proof tamper/refusal/restore.

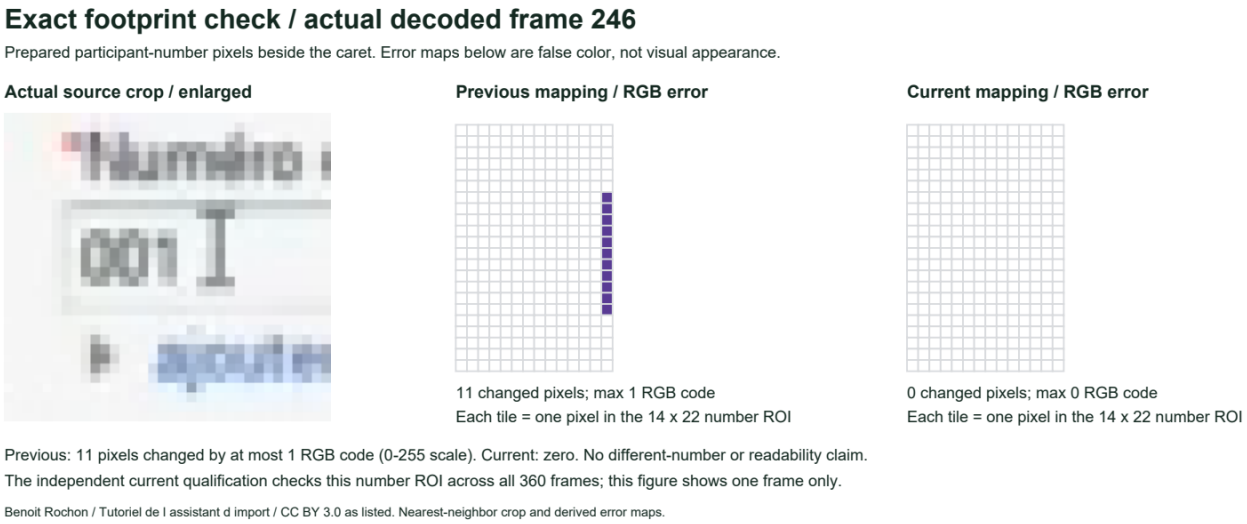
Across these selected 11 excerpts, inside-hold threshold activity totals 50 to0; complete-excerpt activity totals 159 to97, with zero newly timed events. This is the same ROI-mean engineering proxy, not semantic motion, representative recall or a human-benefit result.

This round adds 13 recordings and 91 distinct preselected windows: 18 wiki fixed,24 expanded wiki, 35 GitHub Git Foundations and 14 German Git. One Wikibase excerpt refuses the 16-region interface limit; 90 reach review with exact original/prepared recovery and withheld no-edit exports. Other proposals are not renamed distractions. Same-course recordings are related, and this convenience search is not representative recall. Discovery used recorded v9/pre-luma builds; accepted exports use the final manifest.

The actual app now includes guided Joomla protection: load three editable source-bound areas instead of typing twelve coordinates. Fresh plan and consent remain required. Mobile 375/890 checks use scrollWidth <= clientWidth; a 7px intrinsic-width overflow was fixed. Unfinished marking locks Save/Check/Generate; Cancel restores saved areas. All30 Joomla PNGs pass native HTTP/hash checks. Local browser loading stopped at 12 boundary PNGs when its container exited; complete browser proof/download checks moved to AWS.

A real partial-JSON polling failure led to atomic preparation/error receipt publication. Failure-path tests retain the old receipt and remove temporary files. Original recovery now has a byte-verified .ogv/.webm/.mp4/.mov/.mkv suffix and MIME while preserving its hash-verified bytes. Unknown headers refuse.

Live delivery, reproduction and measured impact



All 14 final Linux stages pass: 137 backend and 113 studio tests (one platform skip). All 117 runtime files are checked. This is an application/example/export layer on the exact v9 parent, not a fresh cold dependency rebuild. OpenCV 5 temporal differences and connected components propose the region; pinned FFmpeg encodes the reviewed sample mapping. Independent decoded gates decide acceptance.

AWS ECS/Fargate default-lesson-rescue:16 completed as one running task. Image sha256:a154dba16c8ad22a50a1ebcf6ac40f2cf9e00304f6d770db9027d479408a1917. Existing 1vCPU/2GiB, one-worker, owner-bound access and resource limits remain. Runtime manifest 20dfbb02970dcfb4d2a8fc0a8414dc900ace3cb16ea75130090cefc12217a8a0.

Actual Brave live verification: Final Brave original French upload, four manual areas and explicit export; all36 static proofs, actual375/890 fit, one complete verified rescue-pack download and reload recovery. Guided Joomla tested separately. No media playback or video changes. Download receipts record actual files and hashes, separate from HTTP-only or historical evidence. No new media playback, recording, narration or submission-video change. The existing video remains historical until the user supplies audio.

At frame 246 the rejected mapping changes 11 adjacent pixels by at most one RGB code; the current mapping changes zero. This shows stricter spatial preservation, not a different typed number or a readability benefit. The recovery pack retains the source/edit/recipe/proofs together before expiry. Time saved, preference, attention, comprehension, accessibility, adoption and safety remain unmeasured; no judge-score prediction is made.

Reproduce: build the root Dockerfile, run python studio/run.py --verify /tmp/qualification, then run the included original-source qualifier against hash-verified sources and exact case JSONs. docs/natural-v10/REPRODUCE.md and evidence/natural-v10 retain selection plans, source credits, failures, recipes, ordered teaching digests and independent measurements. The source archive is a pinned git archive with all 117 runtime file hashes checked and no handoff paths.

Try the live app: choose Joomla installation, inspect both source states, load and inspect suggested teaching areas, Check edit plan, explicitly tick consent, then Generate comparison. Expand every teaching proof and boundary comparison; download wanted results. For the new French original, use start140/length12 and the four case rectangles. Exporting a comparison does not accept it for use.

Limits:1–12 second SDR excerpts, originals <=64MiB. Preparation changes source picture/audio to 640x360/30fps/optional monoAAC once and can reduce legibility. Exact checks reference prepared bytes, not original resolution. No full-lesson splice-back or moving semantic tracking. Jobs expire after 24hours or sooner on restart. A caret may convey meaning; creator review stays required.